

KARTIK SIROHI

kartik.orion.dev@gmail.com | github.com/sirohikartik |
NIT Jalandhar, Punjab, India

EDUCATION

NIT Jalandhar
B.Tech, Information Technology | CGPA: 9.5/10.0

Punjab, India
2024 – Present

EXPERIENCE

AI Research Intern Jan 2026 – Mar 2026
University of Galway Remote

- Co-authored **DICE-RoBERTa** (accepted, ComSys 2026) — a compute-efficient video classification model achieving **85.0% accuracy on UCF101 at only 9.8 GFLOPs/video**, a 5–10× reduction vs. baseline transformers.
- Second paper on neural networks in computational psychiatry currently **under review at IEEE Access**.

Freelance AI / Backend Engineer Aug 2025 – Dec 2025
Independent Remote

- Built **FasTest**, an AI-powered API testing platform with automated test generation and concurrent execution.
- Designed a context-aware LLM pipeline producing structured JSON payloads for edge-case coverage.

PROJECTS

tinyGPT — C++ Transformer Inference Engine | *C++, SIMD (AVX/NEON)* 2026

- Built a **30M parameter decoder-only transformer** from scratch with a custom C++ inference engine.
- Implemented **SIMD-optimized matrix multiplication** (AVX/NEON) and **cache-aware tiled kernels** for constrained environments.
- Engineered localized execution to achieve a **96.9ms Time-to-First-Token (TTFT)** for prompt processing and a generation throughput of **1.88 tokens/sec (537ms/token)** on a baseline consumer Apple M1 Air (8GB RAM).

QuantumSim — Quantum Circuit Simulation Library | *C++* 2025

- Built a modular **single-qubit quantum computing simulation library** in C++ using object-oriented abstractions for quantum states and gate operations.
- Implemented **Hadamard, Pauli-X, Pauli-Y, and Pauli-Z gates** using complex-valued matrix–vector transformations with state normalization validation.
- Designed a **polymorphic gate framework** and command-line circuit visualization system, enabling extensible quantum circuit construction and execution.

Mini-Torch — Neural Network Framework | *Python* 2026

- Built a custom tensor engine with **reverse-mode automatic differentiation** and dynamic computation graph.
- Implemented backpropagation from scratch; SGD and Adam optimizers.

OPEN SOURCE CONTRIBUTIONS

ggml-org/llama.cpp — PR #22209 (Merged) 2026

- Vectorized ggml_vec_dot_q4_1.q8_1** — rewrote hot-path quantized dot product using SIMD intrinsics; directly improves LLM inference throughput.

pytorch/pytorch — PR #172200 (Merged) 2026

- Fixed **NaN propagation bug in tensor clamp operation**; navigated pytorch’s large codebase and strict review process independently.

alexandria3k — PR #63 (Merged) 2026

- Reduced lookup complexity from **O(n)** to **O(1)** via algorithmic restructuring.

scipy/scipy — PR #23129 (Merged) 2025

- Added documentation examples to Gauss-Jacobi quadrature API.

Tracer-Cloud/opensre — PR #852, PR #917, PR #1059 (Merged) 2026

- Added unit tests for data validation and Grafana Tempo mixin in opensre.

TECHNICAL SKILLS

Performance / Systems: SIMD (AVX, NEON), Cache-aware algorithms, Memory optimization, Low-level C++

AI / ML: Transformers, LLMs, CNNs, RAG, Autograd, FAISS, Quantization

Languages: C++, Python, JavaScript

Backend: FastAPI, REST APIs, Async Systems, PostgreSQL

Tools: Git, Linux, Postman, Vercel

Core CS: Data Structures & Algorithms, Operating Systems, OOP, DBMS, Computer System Architecture

ACHIEVEMENTS

- Ranked **1365 / 34,000+** participants in LeetCode Biweekly Contest.
- Secured **3rd place** at HackMol 2025, NIT Jalandhar.